



Northern University

Of Business and Technology Khulna

Course Title: Mobile Computing Lab

Course Code: CSE 4204

Backend Development & Database Implementation

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Date of Submission: 29.06.20

Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence

1.Team Information

Team Name	CSE4204-8B-T04
Section	8B
Project Title	Enterprise Nexus: Multi-Agent AI Business Automation OS
Team Leader	Nazmus Sakib ID: 11220320888
Members	Nazmus Sakib Sabrina Ibrahim Khusbu Shoeb Shariar Masuk Most Shumaiya Shanjidah
Submitted To	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE
Course	CSE 4204 — Mobile Computing Lab
Submission	Week 06 — System Design & System Architecture
Date	29 June 2026
GitHub	https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-SecurityIntelligence

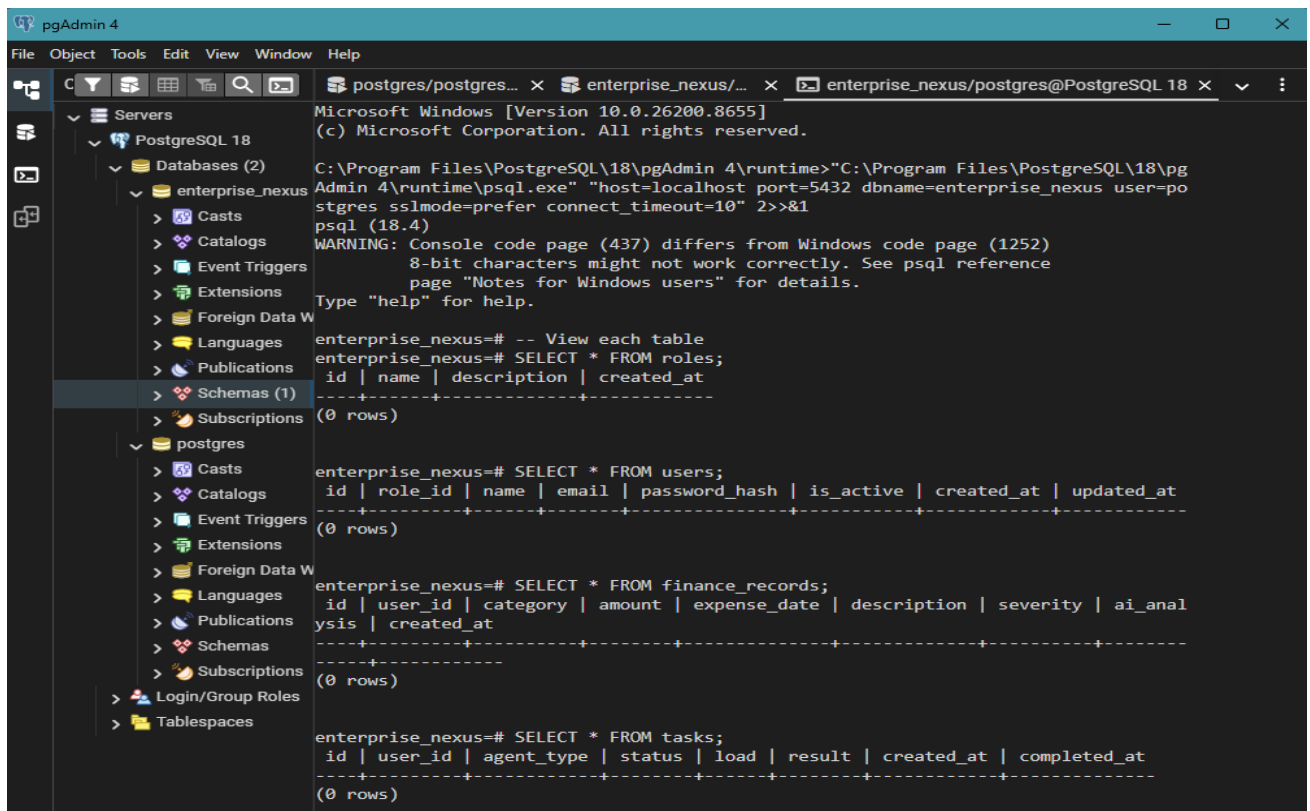
2.Backend Technology Stack

Category	Technology / Tool	Purpose
Runtime Environment	Node.js	Executes JavaScript code on the server.
Framework	Express	Builds RESTful APIs and handles routing and middleware.
Database	PostgreSQL	Stores and manages relational data.
ORM / Database Access	pg (or direct SQL)	Connects to PostgreSQL and executes SQL queries.
Authentication	JWT (JSON Web Token)	Authenticates users and secures API endpoints.
API Style	RESTful API	Provides standard HTTP endpoints (GET, POST, PUT, DELETE).
Containerization	Docker	Packages the application and its dependencies into containers for consistent deployment.
Environment Configuration	dotenv	Loads environment variables from a .env file.
Authentication Middleware	Custom Express Middleware	Verifies JWT tokens and protects private routes.
Input Validation / Sanitization	Middleware (e.g., express-validator or custom)	Validates and sanitizes incoming request data to improve security.
Error Handling	Express Error-Handling Middleware	Catches and processes application errors, returning consistent API responses.

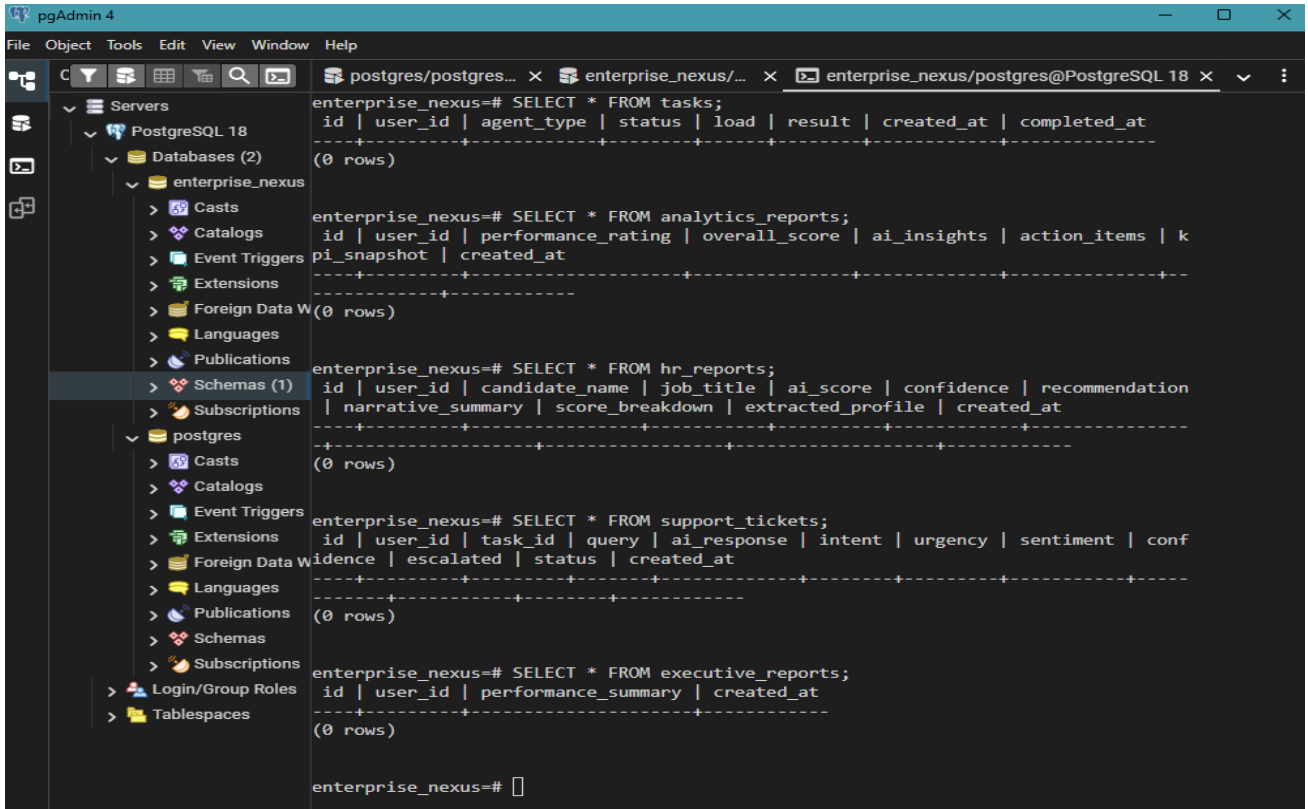
3.Database Design Summary

Database Design Feature	Description
Core Business Modules	The database is organized into core modules, including Users, Roles, Finance, Tasks, Analytics, HR, Support, and Executive Reports , making the system modular and easy to maintain.
Primary Keys	UUID (Universally Unique Identifier) is used as the primary key for all major tables, ensuring globally unique and scalable record identification.
Table Relationships	Relationships are established using foreign keys , such as linking users to roles and connecting domain-specific records to their respective users.
JSONB Data Storage	JSONB columns store flexible, semi-structured data such as task payloads, analytics snapshots, and HR profile information without requiring frequent schema changes.
Database Indexing	Indexes are created on commonly queried columns, including user_id , status , and date fields, to improve query performance and reduce response time.
Automatic Timestamp Updates	A database trigger automatically updates the users.updated_at column whenever a user record is modified, ensuring accurate tracking of changes.
Modular & Scalable Architecture	The schema separates domain-specific data into dedicated tables, making the system easier to extend and supporting future AI-driven features and business growth.

3.1 Database Tables/Collections



Database Table screenshot_1



Database Table screenshot_2

4.Implemented APIs

API Structure and End Point Reference

All endpoints follow RESTful design principles. Base URL: /api/v1. Every protected route requires a valid JWT Bearer token in the Authorization header. Responses follow a consistent JSON envelope: { success, data, message, timestamp }. WebSocket events are emitted on the /ws namespace for realtime dashboard updates.

4.1 AUTH MODULE — Base: /api/v1/auth

Method	Endpoint	Description	Auth	Request Body	Response
POST	/register	Register new user	Public	{ name, email, password, role }	{ token, user }
POST	/login	Authenticate and receive JWT	Public	{ email, password }	{ token, refreshToken, user }
POST	/logout	Invalidate current session	Protected	—	{ message }
GET	/me	Get current user profile	Protected	—	{ user }

Method	Endpoint	Description	Auth	Request Body	Response
PUT	/me	Update profile name and department	Protected	{ name, department }	{ user }
POST	/refresh	Refresh expired JWT	Protected	{ refreshToken }	{ token }
GET	/users	List all users	Admin	?role=&is_active=	{ users[] }
PUT	/users/:id/role	Update user role	Admin	{ role }	{ user }

4.2 HR MODULE — Base: /api/v1/hr

Method	Endpoint	Description	Auth	Request Body	Response
POST	/candidates	Upload CV and create candidate record	HR+Admin	multipart/form-data (cv_file, job_title)	{ candidate }
GET	/candidates	List candidates with filters	HR+Admin	?status=&sort=ai_score&order=desc	{ candidates[], count }
GET	/candidates/:id	Get full candidate detail	HR+Admin	—	{ candidate }
PUT	/candidates/:id	Update status or notes	HR+Admin	{ status, notes }	{ candidate }
DELETE	/candidates/:id	Delete candidate record	Admin	—	{ message }
POST	/candidates/:id/screen	Trigger AI CV screening	HR+Admin	—	{ ai_score, ai_summary, skills }

4.3 FINANCE MODULE — Base: /api/v1/finance

Method	Endpoint	Description	Auth	Request Body	Response
POST	/invoices	Upload invoice for AI processing	Manager +	multipart/form-data (invoice_file)	{ invoice }
GET	/invoices	List invoices	Manager +	?status=&anomaly=true&sort=	{ invoices[], count }
GET	/invoices/:id	Get invoice with AI analysis	Manager +	—	{ invoice }
POST	/invoices/:id/analyze	Re-run AI analysis	Manager +	—	{ analysis }
GET	/anomalies	Get all flagged anomalies	Manager +	?severity=	{ anomalies [] }
GET	/summary	Financial summary by period	Manager +	?period=monthly&year=2026	{ stats, chart_data }
GET	/duplicates	List detected duplicate invoices	Manager +	—	{ duplicates [] }

8.4 SUPPORT MODULE — Base: /api/v1/support

Method	Endpoint	Description	Auth	Request Body	Response
POST	/tickets	Create new support ticket	All Users	{ subject, body }	{ ticket }
GET	/tickets	List tickets with filters	All Users	?status=&priority=&sentiment= =	{ tickets[], count }
GET	/tickets/:id	Get ticket with AI response	All Users	—	{ ticket }
PUT	/tickets/:id	Update status and assignment	Support+	{ status, assigned_to }	{ ticket }
POST	/tickets/:id/ai-respond	Trigger AI autoresponse	Support+	—	{ response }
DELETE	/tickets/:id	Delete ticket (soft delete)	Admin	—	{ message }
GET	/sentiment	Sentiment analytics dashboard data	Manager +	?period=weekly	{ stats, chart_data }

8.5 SECURITY MODULE — Base: /api/v1/security

Method	Endpoint	Description	Auth	Request Body	Response
POST	/scan	Submit content for threat scan	All Users	{ content } or multipart (file)	{ verdict, confidence, event_id }
GET	/events	List security events	Admin	?severity=&type=&from=&to=	{ events[], count }
GET	/events/:id	Get full event detail	Admin	—	{ event }
PUT	/events/:id/quarantine	Quarantine flagged item	Admin	—	{ message }
PUT	/events/:id/release	Release quarantined item	Admin	—	{ message }
GET	/alerts	Live alert feed (WebSocket)	Admin	—	event: security_alert
GET	/stats	Security event statistics	Admin	?period=	{ stats }

4.6 EXECUTIVE MODULE — Base: /api/v1/executive

Method	Endpoint	Description	Auth	Request Body	Response
GET	/dashboard	Real-time aggregated KPI data	Manager+	—	{ kpis, agent_status, alerts }
POST	/reports	Generate AI executive	Manager+	{ period_start, period_end, type, scope[] }	{ report }

Method	Endpoint	Description	Auth	Request Body	Response
GET	/reports	List generated reports	Manager+	?type=&from=&to=	{ reports[] }
GET	/reports/:id	Get full report content	Manager+	—	{ report }
DELETE	/reports/:id	Delete report	Admin	—	{ message }
POST	/query	Natural language business query	Manager+	{ query_text }	{ answer, sources[], confidence }
GET	/analytics	Crossmodule deep analytics	Admin	?module=&period=&metric=	{ analytics }

5.Authentication Workflow

5.1User registration

A new user sends a request to the register endpoint.

The server checks that the name, email, and password are provided.

It verifies that the email is not already registered.

A new account is created with a default role of employee unless another role is provided.

The server returns an access token and the user details.

5.2 User login

The user sends login credentials to the login endpoint.

The system checks whether the email exists and whether the account is active.

It verifies the password against the stored password hash.

If valid, the server issues:

an access token for normal API requests

a refresh token for getting a new access token later

5.3 Accessing protected routes

For protected routes, the client must send the access token in the Authorization header as a Bearer token.

If the token is valid, the user is loaded from the database.

5.4 Role-based authorization

Admin-only actions are restricted to users with the admin role.

Authenticated users can view their own profile.

They can also update basic profile information such as name and department.

When the access token expires, the client can send the refresh token to the refresh endpoint.

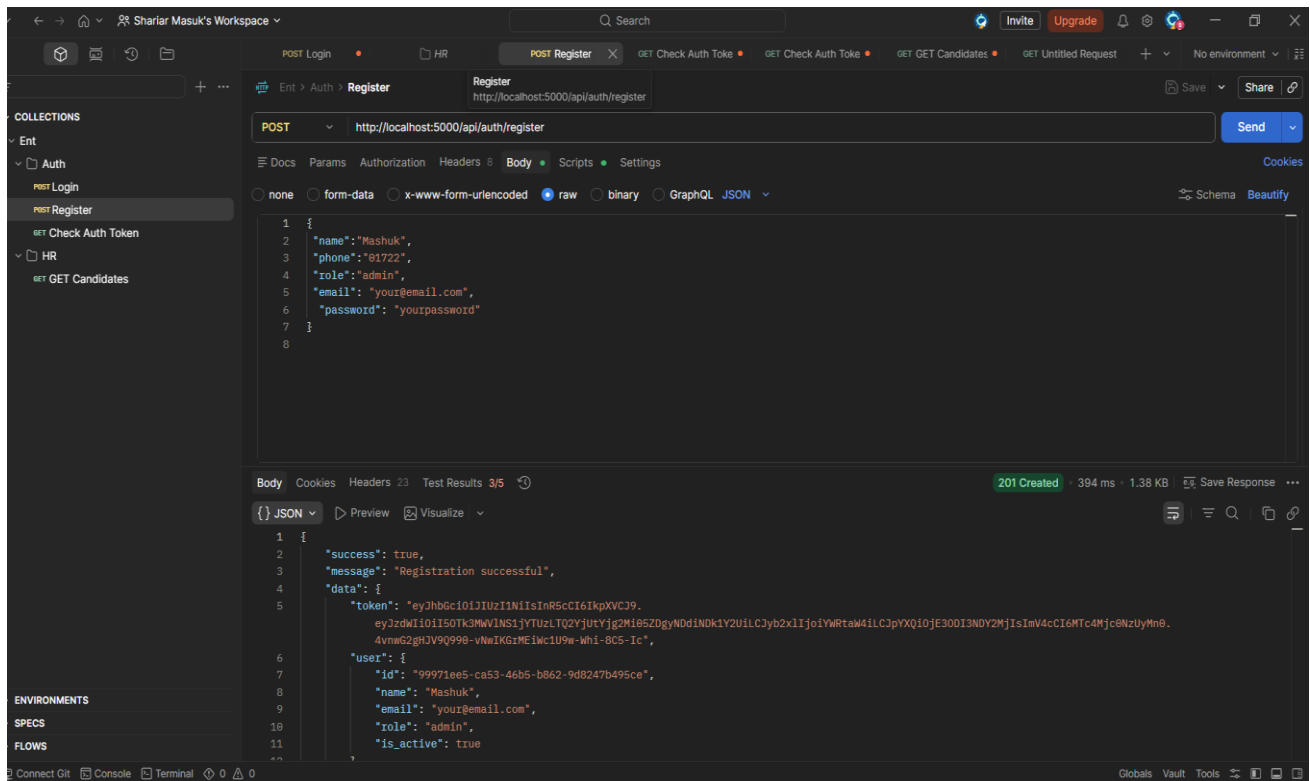
The server validates the refresh token and issues a new access token.

The logout endpoint is available for authenticated users and simply completes the session logout flow.

The project is currently in active development with a backend built around a modular Node.js and Express architecture. Core modules for authentication, user management, finance, HR, analytics, support, and executive workflows have been structured and organized into separate controllers, routes, and agents. Database setup and schema definition are also in progress, with PostgreSQL integration and table creation scripts being prepared. The application is being designed to support AI-driven business operations and modular expansion.

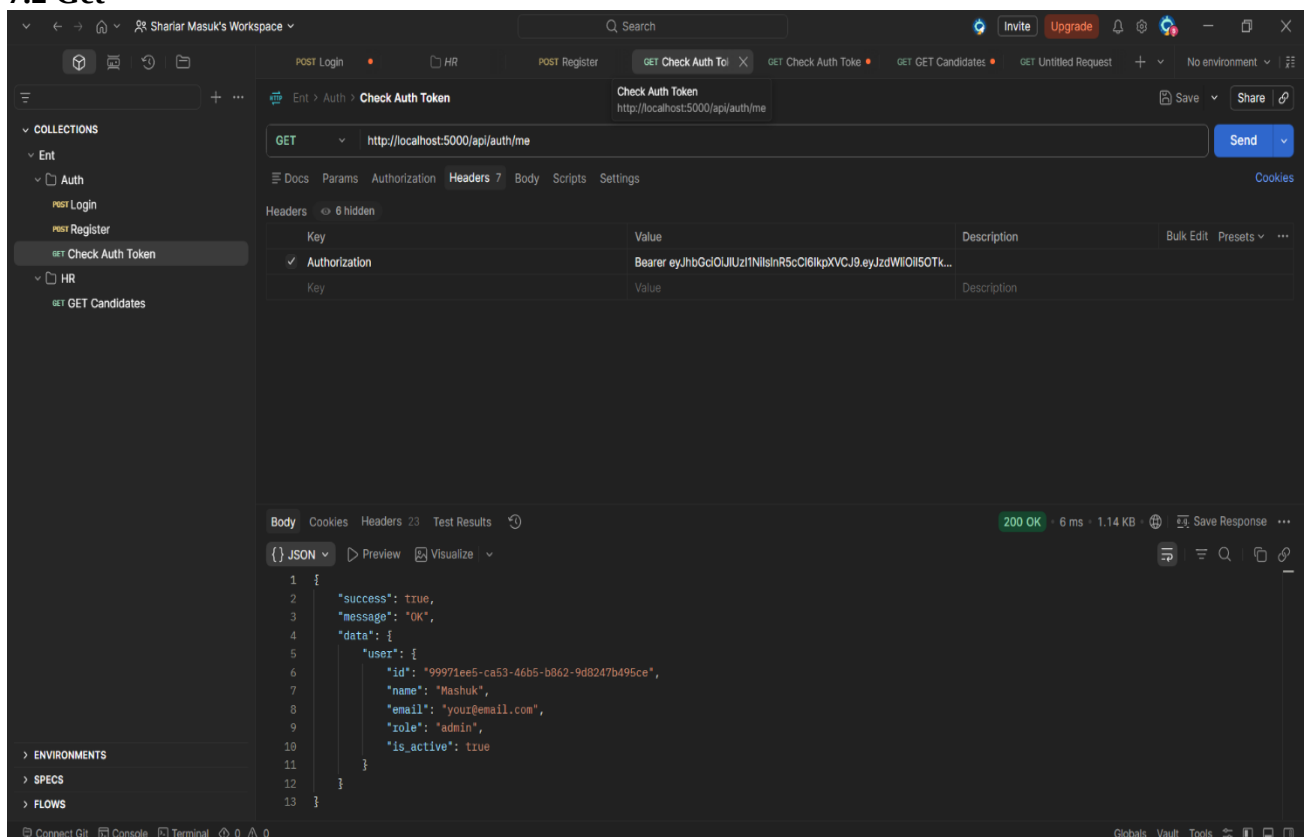
7.1Post



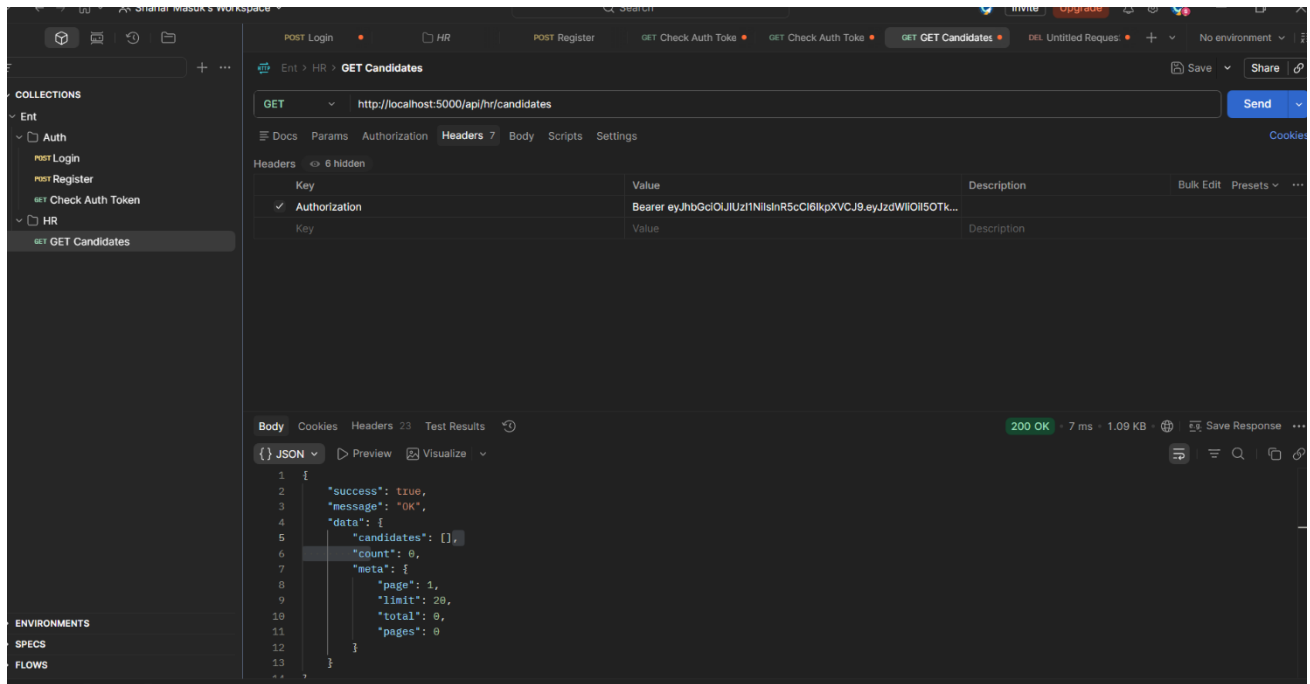


Post_Screenshots_2

7.2 Get

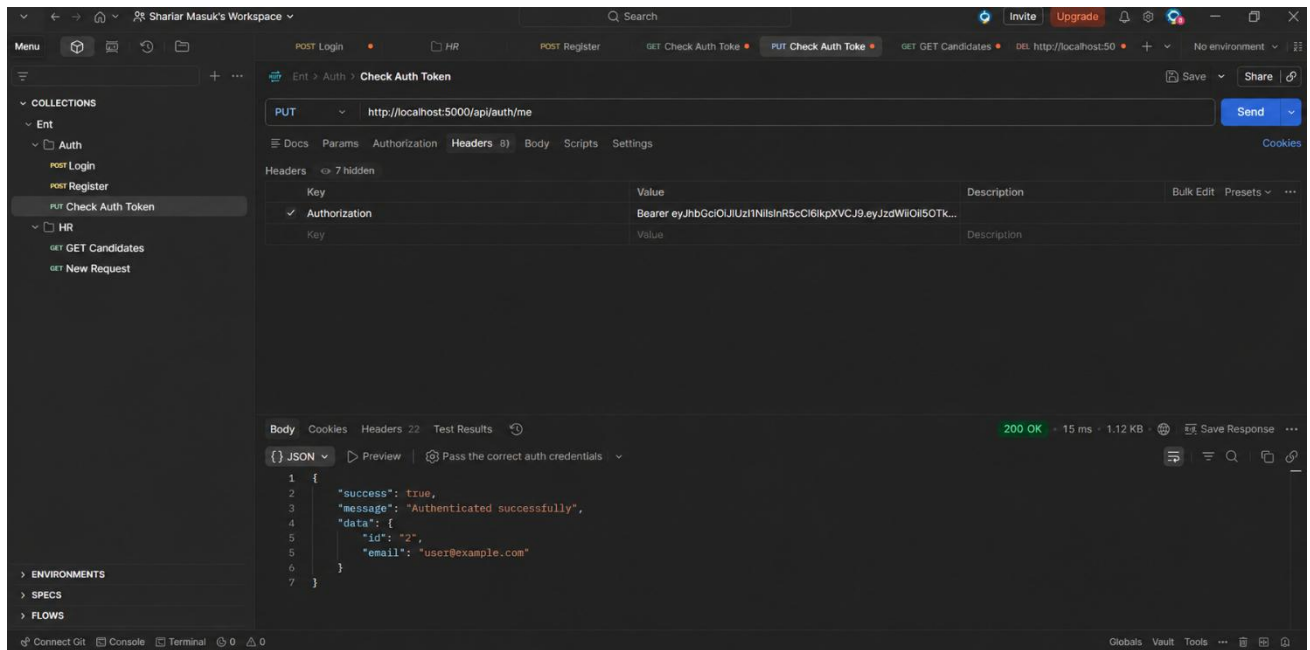


Get-Screenshots-1



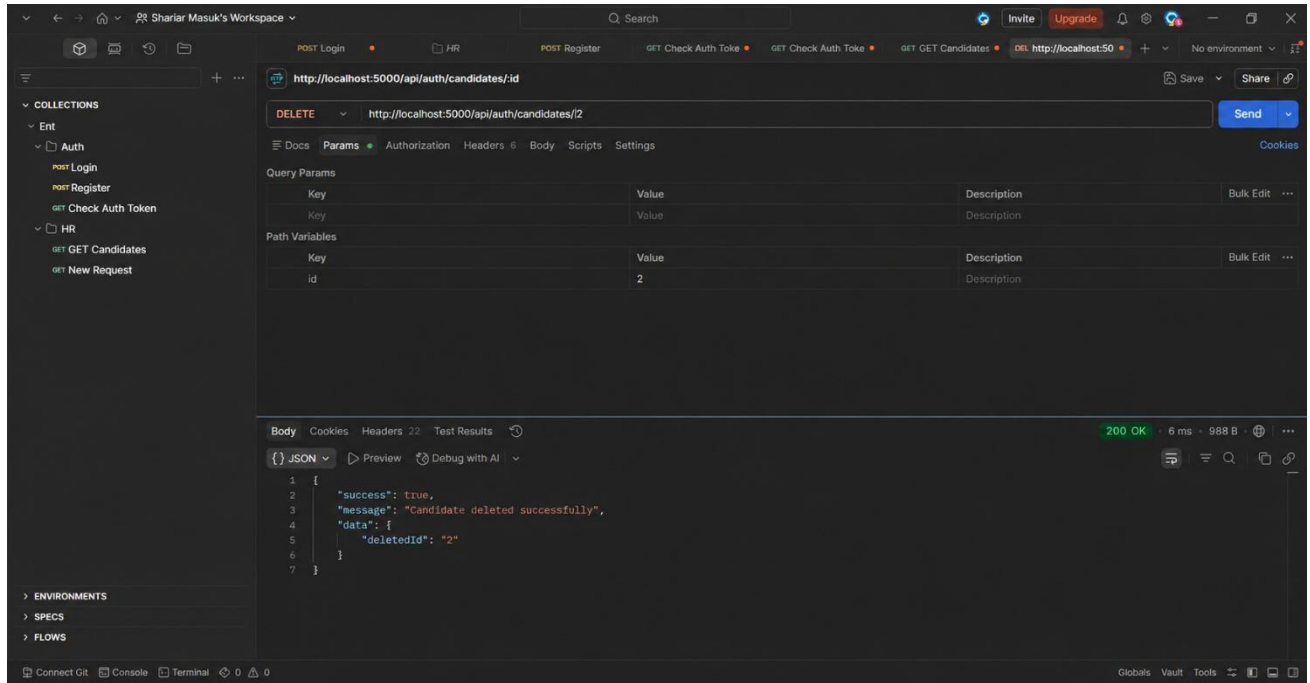
Get-Screenshots-2

7.3 PUT



Put-Screenshots-1

7.4 DELETE



delet-Screenshots-1

8. GitHub Repository Link

<https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence>